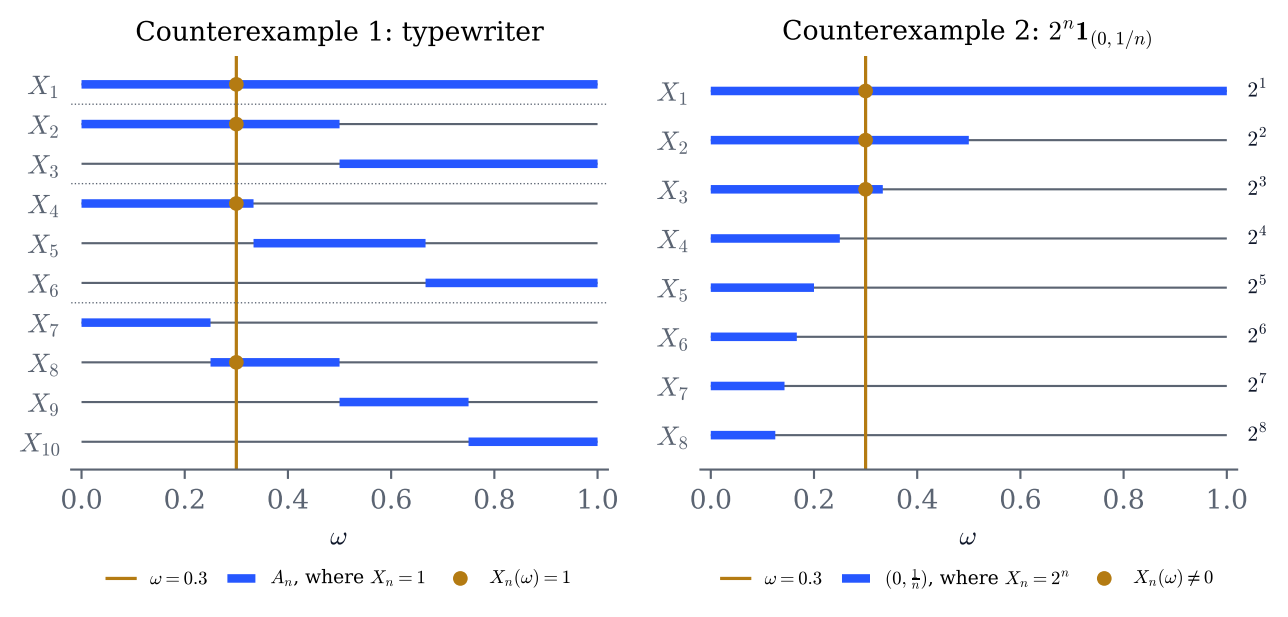
Counterexample 1: typewriter
X_1
X_2
X_3
X_4
X_5
X_6
X_7
X_8
X_9
X_{10}
0.0
0.2
0.4
0.6
0.8
1.0
ω
$\omega = 0.3$
A_n, where $X_n = 1$
$X_n(\omega) = 1$
Counterexample 2: $2^n \mathbf{1}_{(0,\,1/n)}$
X_1
X_2
X_3
X_4
X_5
X_6
X_7
X_8
2^1
2^2
2^3
2^4
2^5
2^6
2^7
2^8
0.0
0.2
0.4
0.6
0.8
1.0
ω
$\omega = 0.3$
$(0, \frac{1}{n})$, where $X_n = 2^n$
$X_n(\omega) \neq 0$